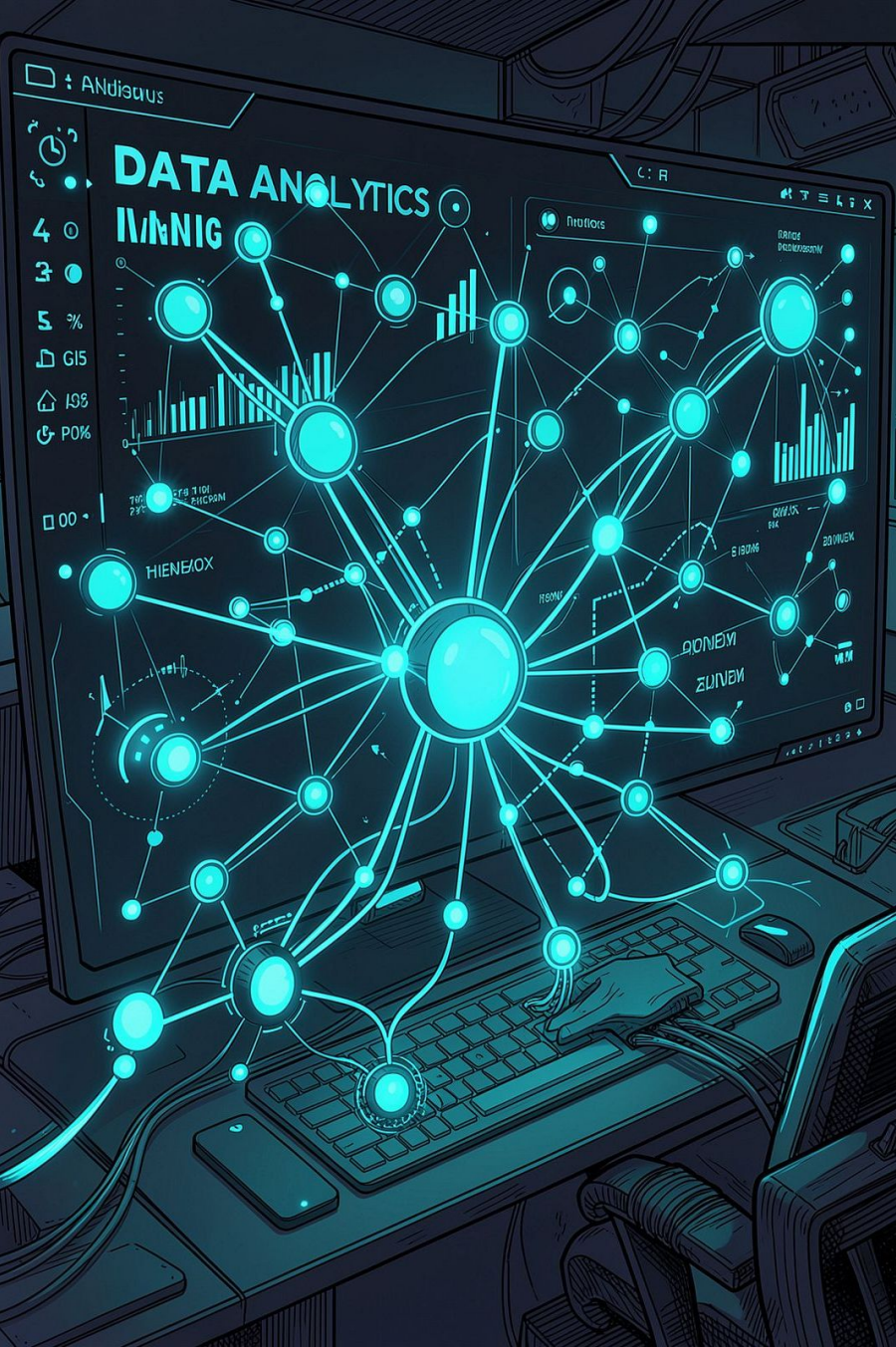




ARCHITECTURE DEEP DIVE

Snowflake + Streamlit: App Architecture

A multi-page analytics application powered by **Snowflake**, **Cortex AI**, and **Streamlit** — covering data connectivity, AI assistance, dashboarding, and export in a single cohesive platform.

Multi-Page Routing & State Management

Dashboard

AI Assistant

Dashboard Builder

Published Dashboard

The app uses `st.navigation()` with a hidden default nav and a custom navigation bar for seamless page routing.

`st.navigation()`

Custom nav bar with hidden default navigation for clean UX

`session_state`

Shared constants + per-page namespaced keys for persistent state

`@st.cache_data`

600s TTL for heavy queries, 60s for lightweight lookups

Snowflake Access & Cortex AI Integration

Snowflake Access Patterns

`conn.query()`

Read-only queries against
Snowflake tables — cached
for performance

`conn.session()`

DDL/DML operations requiring
an active session context

Cortex AI Modes

`CORTEX.COMPL
ETE`

Llama 3.1 70B for ad-hoc
schema generation and
natural-language queries

`CORTEX.DATA_
AGENT_RUN`

Pre-built semantic model for
logistics-domain reasoning

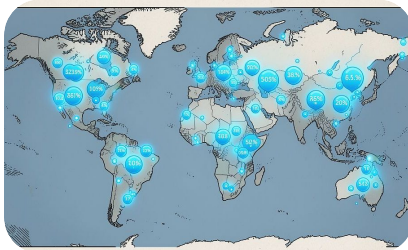
FRONTEND UX

Visualization & Conversational Interface



Vega-Lite Charts

Bar, area, and donut charts rendered declaratively — fully interactive with Streamlit's native support



PyDeck Geospatial Map

Revenue bubble map overlaid on a world map, enabling regional drill-down and spatial analysis

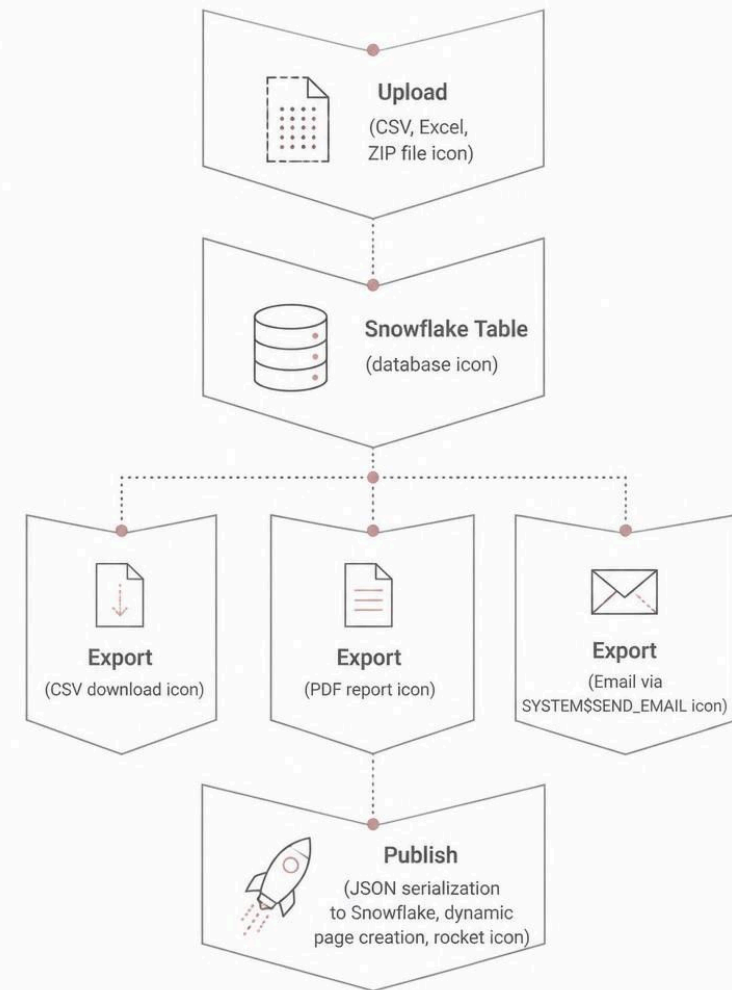


Chat UI with Follow-Ups

Regex-parsed AI responses surface clickable follow-up buttons, guiding users through conversational data exploration

Ingestion, Export & Dashboard Publishing

The app supports a complete end-to-end data lifecycle — from user uploads through to published, shareable dashboards.



01

Upload

CSV, Excel, or ZIP files ingested into temporary or permanent Snowflake tables

02

Export

Three formats: CSV download, custom PDF report, or email via `SYSTEM$SEND_EMAIL`

03

Publish

Dashboard configs serialized as JSON to Snowflake, dynamically creating new app pages